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PI Name:	Hlatky, Lynn Ph.D.		
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PI Email:	Lynn.Hlatky@tufts.edu , LynnHlatky@aol.com	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	(617) 789-2997
Organization Name:	GeneSys Research Institute/Tufts University School of Medicine		
PI Address 1:	736 Cambridge St # CBR-112		
PI Address 2:			
PI Web Page:	http://www.cancer-systems-biology.org		
City:	Boston	**State:**	MA
Zip Code:	02135-2907	**Congressional District:**	8
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
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Task Description:	The goal of our investigation is to provide multi-scale data for an improved mechanistic understanding, and an augmented conceptual framework for NASA's use in estimation of carcinogenesis risk from space radiation. Our approaches, and BNL findings to date, offer mechanistic insight into the radiation modulation of several fundamental "hallmark" processes of cancer that are ubiquitous and extend across tumor types of interest to NASA. It is now recognized that many cancer hallmark processes (including angiogenesis, invasion, and inflammation - a research emphasis specific to this solicitation) involve intercellular interactions, particularly during the promotion and progression phases of carcinogenesis. These ubiquitous, yet understudied, cell interactions play a pivotal role and are rate-limiting steps in the development of all cancers, including radiation-induced cancers. As such, these "bottleneck" processes have a major influence on the risk of radiation-induced carcinogenesis for astronauts, including the risk that radiation-initiated cells develop into cancer, and that radiation can stimulate pervasive, in situ dormant cancer cells to develop to manifest disease. Over the last decade much has been learned about the detailed biology of these critical cancer processes, but little is known about how radiation, particularly space radiation, modulates this cancer biology. Additionally, a deeper understanding of the radiobiology of these fundamental biological processes (e.g. angiogenesis, invasion, inflammation response) will be informative to NASA with regard to more generalized risk concerns, e.g. cardiovascular, or CNS. Our approach continues to be on systems-level investigations (molecular characterization, epigenetics, cell, tissue, and in vivo modulation) defining functional/molecular signatures that characterize the radiation modulation of hallmark processes and the elucidation and linking of different cancer hallmark processes under space radiation action. The mechanistic studies concentrate and build on our team's expertise and BNL data in angiogenesis, invasion, proliferation, and the "seventh hallmark", inflammation, and the radiation response and modulation of progression by mesenchymal stem cells. We have considerable early data on the modulation of these processes (at the RNA transcriptome, microRNA, protein, and cell-levels, along with in vitro functional assays, both in vitro and in vivo) by iron, proton, and gamma radiation. Under this study we will expand, further validate, and knit together these complex multi-scale BNL data by examination of lower doses and dose rates (including silicon ion radiation), further mechanistic epigenetic characterization (including methylation state), and emphasis on in vivo experiments of the radiation-induced systems response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studies to gain an improved mechanistic understanding of carcinogenesis risk from space radiation stand to inform terrestrial experience, and vice versa. The search for important correlates employing our multi-scale, systems-wide approach has benefitted from this premise, including lessons learned from study of, for instance, radiation-induced blood cancers, and more generally the collateral effects radiation has on host tissues that would support tumor growth. The latter include a remarkable radiation-dependence in the inducement of host changes modulating tumor progression. The results of our study of blood cell disorders, comparing radiation- to drug-induced AML, CML, and MDS, confirmed key hypotheses relating to the chromosome damage driven origin of each. To summarize, results supported three tenets: 1) exposure to sparsely ionizing radiation increases frequencies of AML, MDS, and CML comparably; 2) exposure to DNA-damaging drugs, especially alkylators, increases frequencies of AML and MDS far more than it increases the frequency of CML. The difference in increased frequencies is 100- to 1000-fold. We reasoned this is because sparsely ionizing radiation is a relatively powerful inducer of balanced reciprocal translocations which are stably transmissible to daughter cells and which favors causing CML. In contrast, DNA-damaging drugs preferentially produce more complex, less stably transmissible mutations, which favors causing AML and MDS. In the study of radiation effects on host support for tumor growth, it was found 56Fe ions promoted tumor angiogenesis and other intercellular determinants of carcinogenesis risk, while protons generally inhibited these processes. More surprisingly, gamma radiation was found to evoke host responses more similar in general direction to 56Fe than protons, despite the physical similarities of action between proton and gamma radiations, as reflected by their very similar RBEs. Both studies offer lessons that affirm and refine hypotheses of terrestrial radiation action. The blood cell disorder studies affirm a chromosomal mechanism for a class of pre-neoplastic and neoplastic blood diseases, with a bias to simpler aberrations in the radiation case showing a greater propensity to induce CML. The favoring of AML and MDS in the drug case gives grounding to the observations that 1) these are rooted in more complex aberrations, and 2) about 30% of MDS patients progress to AML. Further, the effects of radiation on host support is much more dependent on radiation type than previously thought; extending even to comparison between gamma and protons. It was heretofore thought that the comparable physics of final action of protons and gamma for endpoints like cell death would extend to other biological endpoints, but we now know this is not true. Implications extend to both carcinogenesis risk from ambient exposures, as well as to therapeutic exploitation.
	Through a multi-scale approach, we have sought to advance our understanding of carcinogenesis risk. In this approach, we recognize that cancer evolves not only due to events at the molecular scale as part of initiation, but include substantial modulatory cell-cell and tumor-host interactions as cancer cells become promoted and progress to frank disease. --Molecular Scale Studies at Initiation-- Exploring the beginning of this spectrum, we considered epidemiological and clinical data to form the hypothesis that exposure to sparsely ionizing radiation favors development of chronic myelogenous leukemia (CML) and some forms of acute myeloid leukemia (AML) and myelodysplastic syndrome (MDS) that are associated with synchronous proximal DSB lesions (balanced reciprocal translocations and interstitial deletions and inversions). This contrasts with exposure to DNA-damaging drugs, which pre-disposes to other types of mutations and causes AML and MDS but not chronic phase CML. For the hypothesis to prevail, we needed to explain certain features of development of CML, AML, and MDS. Ionizing radiation is comparatively effective in producing balanced reciprocal translocations such as t(9;22) which results in BCR/ABL1 and causes chronic-phase CML. In contrast, many DNA-damaging drugs, especially alkylators, produce other kinds of small- and large-scale DNA damage. These drugs rarely produce spatially proximal DSB pairs that might create smaller deletions but some are effective at producing isolated DSBs that can lead to chromosome arm losses (terminal deletions) when the free ends of a single DSB fail to rejoin. This means exposure to DNA-damaging drugs

Task Progress:

may kill cells with BCR/ABL1 before CML is manifest and concurrently may cause MDS or AML through the somatic cell evolution mechanisms discussed above. The predicted consequence is an excess of AML and MDS after exposure to DNA-damaging drugs compared to CML. To explain the linear quadratic risk of AML in the A-bomb survivors versus the linear risk of CML, we suggested the linear CML risk results from tethering between ABL1 on chromosome 9 and BCR on chromosome 22 in the interphase nucleus. This spatial proximity favors a translocation between these genes because if both loci acquire DSBs these breaks are proximal and more likely to incorrectly rejoin. Such proximity effects are particularly strong for ionizing-radiation-induced DSBs since ionizing events of the same particle track occur in synchronous clusters. Tethering between BCR and ABL1 increases drug- and radiation-induced probabilities of t(9,22), but it increases radiation-induced probabilities by a substantial additional factor. Whether interphase tethering underlies the genes involved in AML subtypes with balanced translocations such as t(8;21) and t(15;17) (ETO [RUNX1T1]/RUNX1 [AML1] and PML/RAR-alpha) is unknown. MDS risk was also linear in the A-bomb survivors, suggesting there may also be tethering between loci for some MDS-causing genes in the interphase nucleus. Obvious candidates would be two loci on 5q or7q since del(5q) and del(7q) lesions are more characteristic of MDS than of de novo AML.

If our hypothesis is correct, one might expect an increasing frequency of AML and MDS with combinations of DNA-damaging drugs and sparsely ionizing radiation. This is especially so if different types of mutations are needed to produce these cancers and if DNA-damaging drugs and sparsely ionizing radiation are differentially efficient at producing them. The result could be deleterious synergy. To the extent the complex aberrations associated with alkylating agents is more emulative of space radiations than sparsely-ionizing radiations, one might expect a similar trend.

--Studies of cell-cell interactions in carcinogenesis -- the case for 56Fe--

In the assessment of radiogenic cancer risk from space flight, it is imperative to consider effects not only on the creation of cancer cells (initiation) but also on cell-cell interactions that play an important and often decisive role in the promotion and progression phases. Autopsy results confirm that most adults carry fully malignant tumors that are held in check at a small size and will never become symptomatic. This introduces the possibility that cosmic radiation may significantly influence cancer risk through alteration of the bottleneck inter-tissue interactions responsible for maintaining this dormant state. One such bottleneck is the growth limitation imposed by the failure of the tumor to induce blood vessels (angiogenesis). Other deciding events are the ability of a tumor to proliferate and invade. We have shown that proton radiation, the most prevalent radiation in space, has a suppressive effect on all three of these functional responses. It downregulates angiogenic genes like VEGF and HIF-1a and impairs cell invasion and tumor growth. We decided to test these responses after 56Fe irradiation, an HZE radiation type present in the cosmic environment with presumably high carcinogenic potential.

Human microvascular endothelial cells (HMVEC) and normal human dermal fibroblast (NHDF) cells were irradiated with different doses of 56Fe ion radiation (1GeV/n) at Brookhaven National Laboratory and RNA was extracted 6h later. Genomic-wide array analysis was done on the isolated RNA through the Agilent Platform. It was observed that several pro-angiogenic genes like VEGF, IL-6, and HIF-1a were significantly upregulated after treatment with 56Fe ion radiation. These results were also confirmed at the mRNA and protein levels with the human and murine lung cancer lines, A549 and LLC, respectively. Additional verification of modulation of these key genes was also observed when lungs of C57BL/6 mice treated with 56Fe ion radiation showed an increase in VEGF and MMP9 mRNA and protein expression 6h post-irradiation. Cell invasion was shown to be increased by 56Fe ion radiation in various cell types, including fibroblast, tumor, and endothelial progenitor cells. 56Fe ion irradiation also modulated functional processes crucial to angiogenesis. It enhanced the ability of untargeted (bystander) endothelial cells to invade and proliferate in response to factors produced by targeted fibroblast or cancer cells in vitro. Results also carry over to in vivo. C57BL/6 mice exposed to whole-body irradiation with 0.2Gy dose of 56Fe and injected subcutaneously with LLC tumor cells showed a significant augmentation in tumor growth and growth rate in the irradiated group. Additionally, nude mice exposed to whole-body 56Fe radiation and injected intravenously with A549 cancer cells 3h post-irradiation demonstrated a significant enhancement in lung colonization capacity when compared with the sham-irradiated control mice injected. These results together suggest cell and tissue-level responses to 56Fe irradiation may act to overcome major cancer progression-level bottlenecks including those related to angiogenesis, cell proliferation, and invasion. This is of significant concern for cancer risk estimations pertinent to NASA as achieving these cancer hallmark processes can make the difference between a radiation-induced cancer cell progressing to a clinically detectable cancer in astronauts or not.

To summarize, we demonstrated a strong pro-tumor effect for 56Fe (1GeV/n), manifested through influences on intercellular interactions in the progression phase of carcinogenesis.

--Studies of cell-cell interactions in carcinogenesis -- the case for space protons--

Although the physical aspects of proton radiobiology are well understood, biological endpoints are understudied. In particular, the current practice of assuming the RBE of low LET protons to be a generic value of about 1.1 relative to photons, gained often from direct cell killing or other direct cell-targeting endpoints, likely obscures important unrecognized differentials in induced responses in adjacent tissues that may secondarily influence the radioresponse of target cells. This could be especially important in carcinogenesis, as evidenced by the previously mentioned autopsy findings of lesions in most people dying from non-cancer causes. A deeper understanding of the biological properties induced by proton radiation in the host setting thus stands to inform carcinogenesis, particularly in the later stages of cancer development where tumor-host interactions are most critical. Our recent findings suggest that low-LET proton radiation may exert tumor-suppressive effects through the host tissue, contrary to gamma radiation, which works more to facilitate host support for tumor growth. This striking finding stands in marked contrast to the closely paralleling actions of protons and gamma assumed based on physical considerations and classic RBE endpoints alone.

It now appears that protons induce their own distinctive biological response signature, notably in the endpoints of angiogenesis, cell invasion, and cancer metastasis. Tumor-angiogenesis is essential for both the growth and malignant progression of all tumors. Angiogenesis, in any of its manifestations, is driven by the balance between angiogenic stimulators, e.g., vascular endothelial growth factor (VEGF) or interleukin-8 (IL-8), which promote neovascularization and endogenous angiogenesis inhibitors (e.g., thrombospondin) which inhibit neovascularization. In the case of cancer, angiogenic factors secreted from cancer cells induce the homing of endothelial cells establishing neo-vasculature that feeds the expanding tumor mass. Circulating angiogenic factors have been correlated to tumor progression and poor patient prognosis. We have shown that key pro-angiogenic factors such as VEGF, IL-8, IL-6, and HIF-1a, which shift

the balance towards a pro-angiogenic state, are generally upregulated by gamma radiation, a trend that has been observed in other studies. By contrast, our studies at BNL using 0.5-2Gy dose, 1GeV protons show these endpoints to be downregulated by proton radiation in human dermal fibroblast, endothelial, and A549 carcinoma cells. VEGF concentration in blood of C57BL/6 mice exposed to 2Gy of proton radiation (250MeV) was significantly decreased 21 days post-irradiation compared to either 0Gy controls or 2Gy gamma irradiated mice. We also showed that human A549 lung tumor cells irradiated with 0Gy or 2Gy of either proton (1GeV) or gamma, and then injected after 3-6hrs subcutaneously in mice, grew tumors which exhibited differentials in neovascularization, with reduced angiogenesis in the proton-irradiated group, as detected by CD31 staining.

Like tumor angiogenesis, tumor cell invasive capacity is also an important hallmark process in carcinogenesis and particularly in cancer progression and spread. Both invasion and metastasis are cancer progression phase processes that impact patient outcome. Given the potential of biological modulation and suppression of core carcinogenesis processes by proton radiation exposure, specific customized radiation exposures could be considered as a means, not only of local tumor control through cell killing, but of inhibition of cancer metastatic spread, via suppression of critical biological progression-level processes that tie to metastasis.

While gamma radiation has been shown by a number of investigators to increase cell invasion and migration of both endothelial and tumor cells, proton radiation, on the other hand, appears to act differently with respect to these processes. We showed that proton irradiation (1GeV) significantly decreased cell migration and the invasive capabilities of fibroblast, endothelial, and A549 tumor cells at a dose of 2Gy. In addition, we observed that primary endothelial cells co-cultured in transwells with proton-irradiated lung adenocarcinoma (A549) or proton-irradiated primary fibroblast cells has reduced invasive capacity through a matrigel-coated membranes, as compared to those endothelial cells co-cultured with sham-irradiated controls. This is indicative of proton-induced paracrine signaling originating from proton-irradiated A549 carcinoma cells or proton-irradiated fibroblasts, acting on the un-irradiated endothelial cells, inhibit endothelial cell invasive capacity. The addition of recombinant IL-8 or VEGF was seen to partially rescue the anti-invasive effects of proton irradiation, demonstrating that proton-induced inhibition of IL-8 and of VEGF directly contributes to the suppression of cell invasion. We showed that miR-200c, a micro-RNA that is an established and prominent inhibitor of epithelial-mesenchymal transition (EMT), a process strongly linked to enhanced invasion and metastasis, was upregulated by proton irradiation (2Gy, 1GeV) in a variety of cell types investigated (fibroblast, endothelial, and tumor). Our preliminary results also show that inhibition of miR-200c in human adenocarcinoma A549 cells abrogates the reduced cell invasion observed post proton irradiation.

Our results also showed inhibition in lung colonization capacity of proton-irradiated A549 cells. In this case, animals were injected with proton-irradiated A549 cells, which had been irradiated with a single dose of 2Gy and then injected intravenously into mice via tail-vein injections. A substantial reduction in the number of tumor nodules was observed in the animals injected with irradiated cells as compared with those injected with control, sham irradiated cells. The result is not explained by the cell killing from the pre-irradiation of the A549 cells. Suggested is a decreased ability of the surviving tumor cells to draw support from the host environment to facilitate tumor growth.

In conclusion, high-energy proton radiation my have carcinogenesis progression-limiting effects at a number of levels, including reduced angiogenesis, decreased cell invasiveness, and lowered colonization efficiency, attributes suggesting reduced tumor progression.

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